

C PROGRAMMING :FIRST FIT

```
GNU nano 9.0 ff.c *
#include <stdio.h>
int main(){
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb, np, i, j;
    printf("Enter Number of Blocks: ");scanf("%d",&nb);
    printf("Enter Number of Processes: ");scanf("%d",&np);
    printf("Enter Block Sizes:\n");
    for(i=0;i<nb;i++){
        scanf("%d",&blockSize[i]);
        printf("Enter Process Sizes:\n");
        for(i=0;i<np;i++){
            scanf("%d",&processSize[i]);
            for(i=0;i<nb;i++){
                allocation[i]=-1;
                for(j=0;j<np;j++){
                    if(blockSize[j] >= processSize[i]){
                        allocation[i]=j;
                        blockSize[j]-=processSize[i];
                        break;
                    }
                }
                printf("\nProcess No\tProcess Size\tBlock No\n");
                for(i=0;i<np;i++){
                    printf("%d\t%d\t%d\t\t",i+1,processSize[i]);
                    if(allocation[i]!=-1)
                        printf("%d\n",allocation[i]+1);
                    else
                        printf("Not Allocated\n");
                }
            }
        }
    }
    return 0;
}
```

OUTPUT :

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano ff.c
(kali@YUVARAJK)-[~]
$ gcc ff.c -o ff
(kali@YUVARAJK)-[~]
$ ./ff
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No      Process Size    Block No
1               2               1
2               2               2
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMINNG : FIRST FIT



A screenshot of a terminal window with the nano text editor open. The editor is editing a file named 'ff.sh'. The script content is as follows:

```
GNU nano 9.0 ff.sh
#!/bin/bash
echo "First Fit Memory Allocation Demonstration"

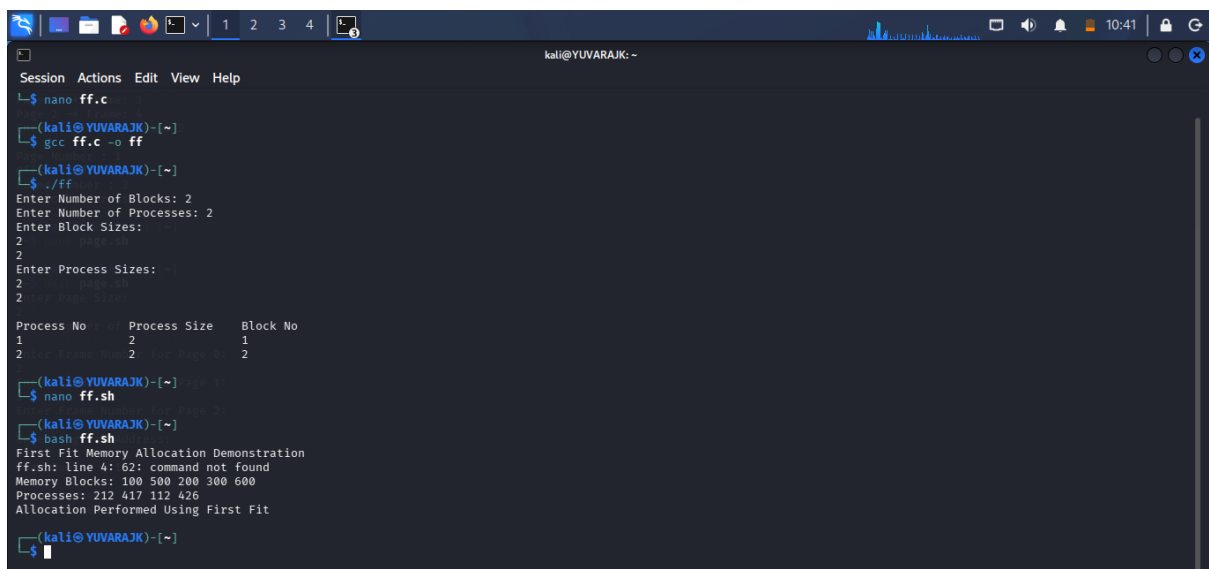
62

blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using First Fit"

# Enter Number of Blocks
# Enter Page Size
# Enter Number of Pages
# Enter Frame Number for Page 0
# Enter Frame Number for Page 1
# Enter Frame Number for Page 2
# Enter Logical Size
# Page Number
# Enter Page Size
# Enter Number of Pages
```

The bottom of the screen shows the nano editor's command palette with various shortcuts like 'G Help', 'X Exit', 'O Write Out', 'R Read File', 'I Where Is', 'A Replace', 'C Cut', 'P Paste', 'E Execute', 'J Justify', 'L Location', 'G Go To Line', 'U Undo', 'R Redo', 'A Set Mark', 'B Copy', 'I To Bracket', 'B Where Was', 'B Previous', and 'F Next'.

OUTPUT :



A screenshot of a terminal window showing the execution of the 'ff.sh' script. The user enters 'nano ff.c' and 'gcc ff.c -o ff' to compile the script. Then, they run './ff' which prompts for input. The user enters 2 for the number of blocks and 2 for the number of processes. The script then prompts for block sizes and process sizes. The user enters 2 for both. The script then displays a table of process sizes and block numbers.

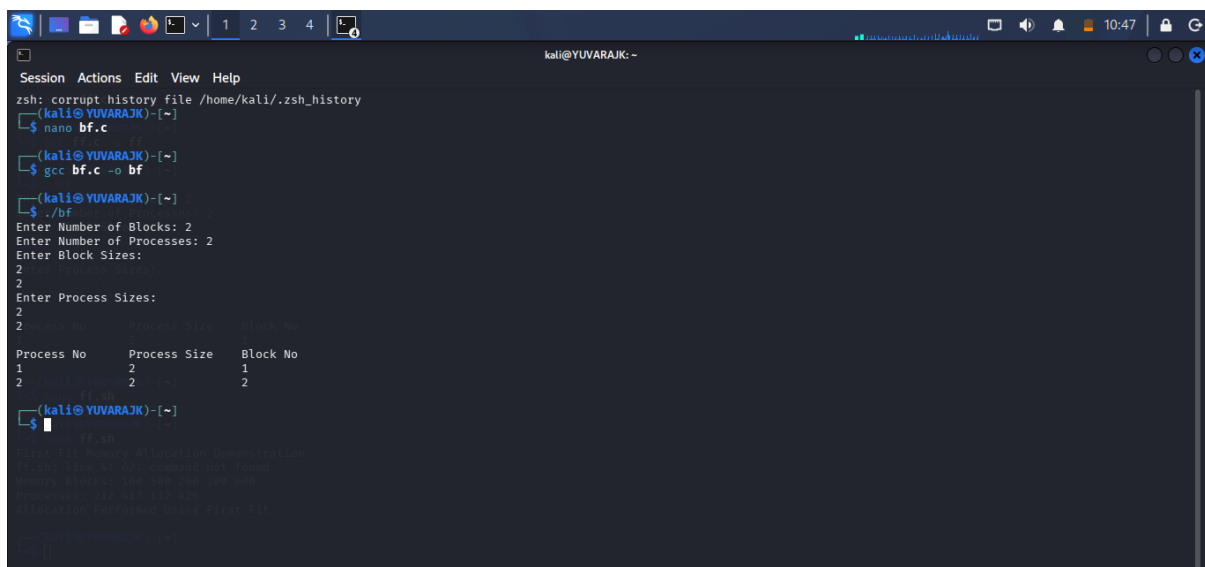
```
(kali@YUVARAJK)-[~]
$ nano ff.c
(kali@YUVARAJK)-[~]
$ gcc ff.c -o ff
(kali@YUVARAJK)-[~]
$ ./ff
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No    Process Size    Block No
1             2             1
2             2             2
(kali@YUVARAJK)-[~]
$ nano ff.sh
(kali@YUVARAJK)-[~]
$ bash ff.sh
First Fit Memory Allocation Demonstration
ff.sh: line 4: 62: command not found
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using First Fit
(kali@YUVARAJK)-[~]
$
```

C PROGRAMMING : BEST FIT



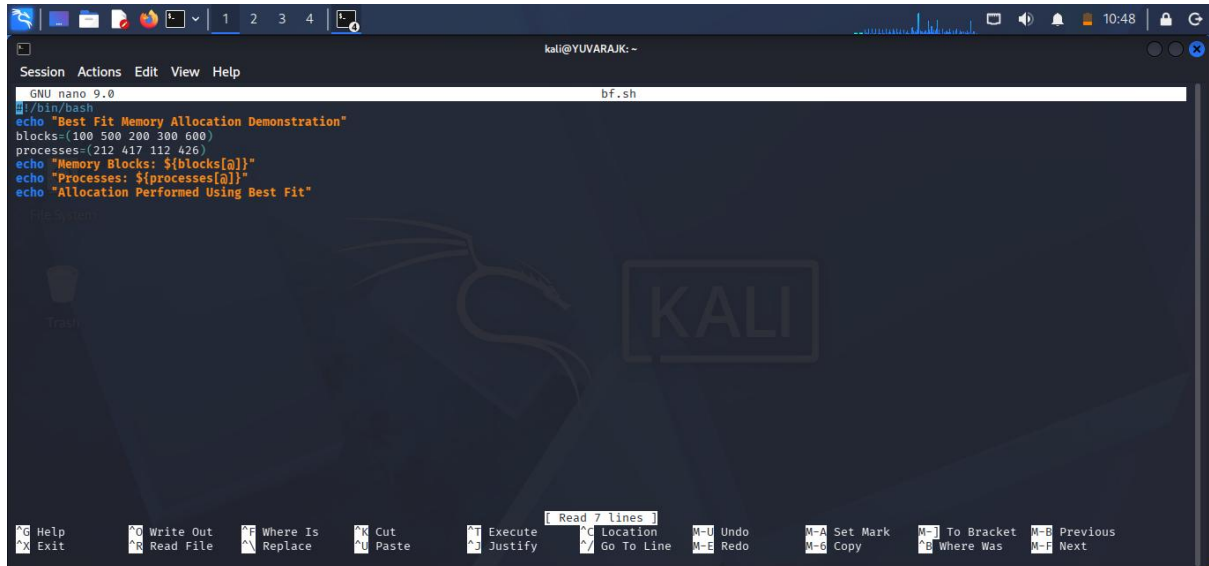
```
GNU nano 9.0 bf.c *
#include <stdio.h>
int main()
{
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb,np,i,j,bestIdx;
    printf("Enter Number of Blocks: ");scanf("%d",&nb);
    printf("Enter Number of Processes: ");scanf("%d",&np);
    printf("Enter Block Sizes:\n");
    for(i=0;i<nb;i++)
        scanf("%d",&blockSize[i]);printf("Enter Process Sizes:\n");
    for(i=0;i<np;i++)
        scanf("%d",&processSize[i]);
    for(i=0;i<np;i++){allocation[i]=-1;
        for(j=0;j<nb;j++){
            if(blockSize[j] >= processSize[i]){
                if(bestIdx== -1 || blockSize[j] < blockSize[bestIdx])bestIdx=j;
            }
        }
        if(bestIdx != -1){
            allocation[i]=bestIdx;
            blockSize[bestIdx]-=processSize[i];
            printf("\nProcess No\tProcess Size\tBlock No\n");
            for(i=0;i<np;i++){
                printf("%d\t\t%d\t\t",i+1,processSize[i]);
                if(allocation[i] != -1)
                    printf("%d\n",allocation[i]+1);
                else
                    printf("Not Allocated\n");
            }
        }
    }
    return 0;
}
```

OUTPUT :



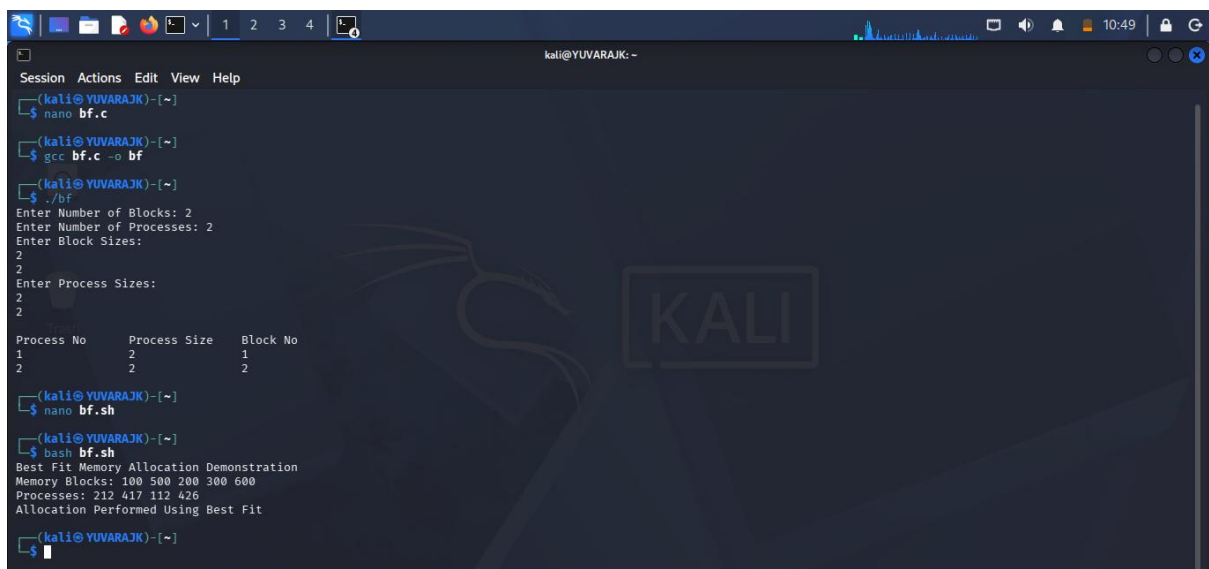
```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano bf.c
(kali@YUVARAJK)-[~]
$ gcc bf.c -o bf
(kali@YUVARAJK)-[~]
$ ./bf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No      Process Size    Block No
1               2               1
2               2               2
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : BEST FIT



```
GNU nano 9.0 bf.sh
#!/bin/bash
echo "Best Fit Memory Allocation Demonstration"
blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using Best Fit"
```

OUTPUT :



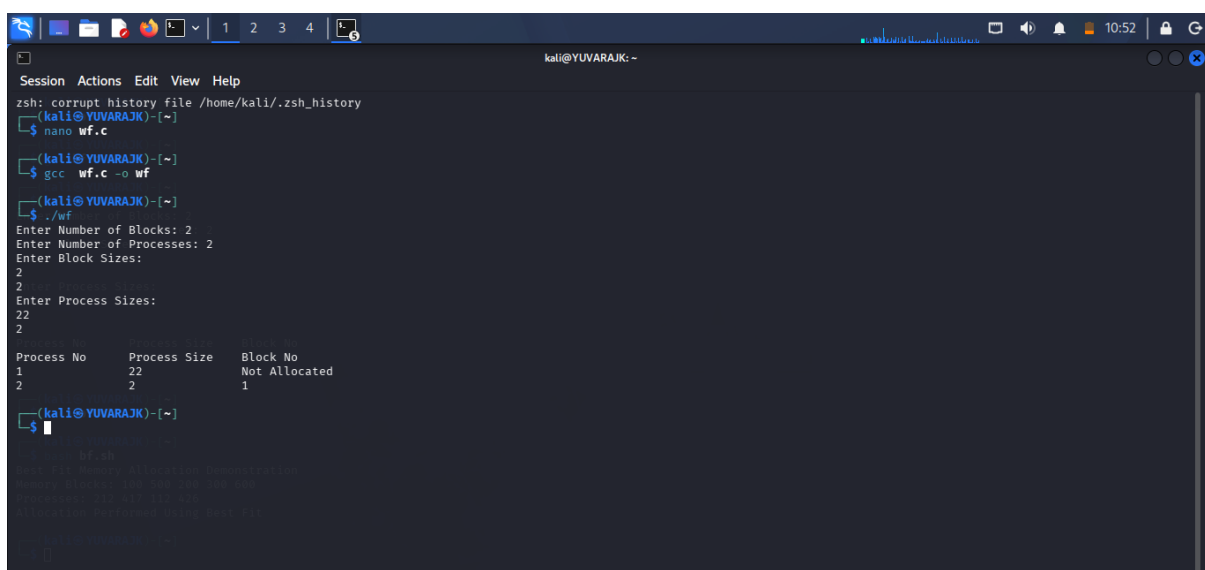
```
(kali@YUVARAJK)-[~]
$ nano bf.c
(kali@YUVARAJK)-[~]
$ gcc bf.c -o bf
(kali@YUVARAJK)-[~]
$ ./bf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No      Process Size    Block No
1                2                1
2                2                2
(kali@YUVARAJK)-[~]
$ nano bf.sh
(kali@YUVARAJK)-[~]
$ bash bf.sh
Best Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using Best Fit
(kali@YUVARAJK)-[~]
$
```

C PROGRAMMING : WORST FIT



```
GNU nano 9.0 wf.c *
#include <stdio.h>
int main(){
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb,np,i,j,worstIdx;
    printf("Enter Number of Blocks: ");scanf("%d",&nb);
    printf("Enter Number of Processes: ");scanf("%d",&np);
    printf("Enter Block Sizes:\n");
    for(i=0;icnb;i++){
        scanf("%d",&blockSize[i]);printf("Enter Process Sizes:\n");
    }
    for(i=0;icnp;i++){
        scanf("%d",&processSize[i]);
    }
    for(i=0;icnp;i++){
        allocation[i]=-1;
        for(j=0;jcnp;j++){
            if(blockSize[j] >= processSize[i]){
                if(worstIdx==-1 || blockSize[j] > blockSize[worstIdx])worstIdx=j;
            }
        }
        if(worstIdx!=-1){
            allocation[i]=worstIdx;
            blockSize[worstIdx]-=processSize[i];
            printf("\nProcess No\tProcess Size\tBlock No\n");
            for(i=0;icnp;i++){
                printf("%d\t\t%d\t\t",i+1,processSize[i]);
            }
            if(allocation[i]!=-1){
                printf("%d\n",allocation[i]+1);
            }
            else{
                printf("Not Allocated\n");
            }
        }
    }
    return 0;
}
```

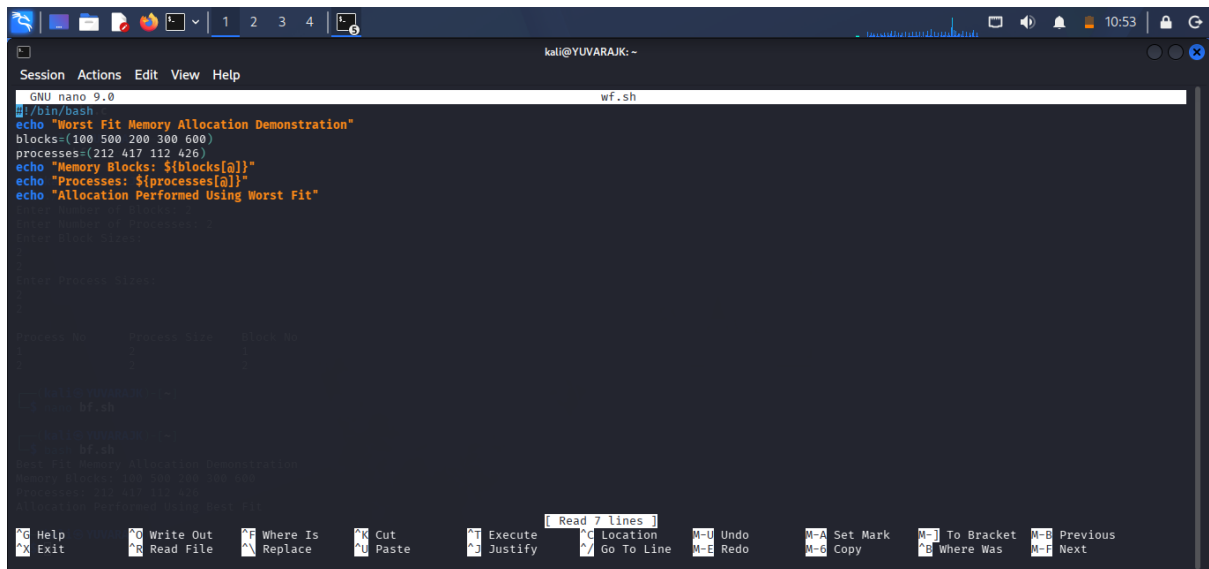
OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano wf.c
(kali@YUVARAJK)-[~]
$ gcc wf.c -o wf
(kali@YUVARAJK)-[~]
$ ./wf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
22
2
Process No      Process Size    Block No
1               22             Not Allocated
2               2              1

(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : WORST FIT



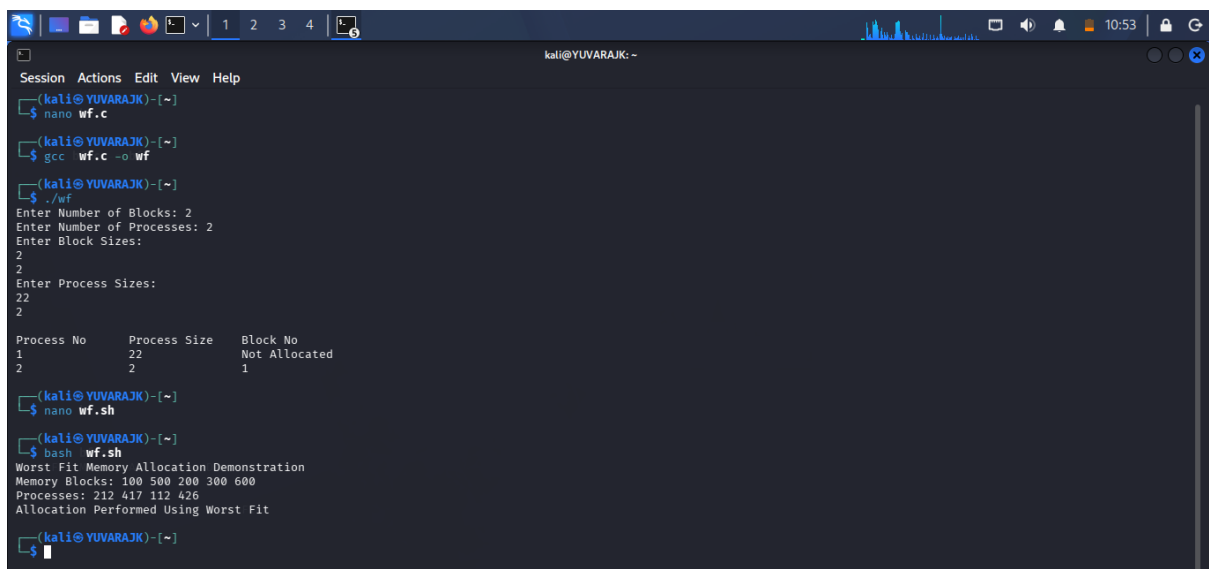
```
GNU nano 9.0 wf.sh
#!/bin/bash
echo "Worst Fit Memory Allocation Demonstration"
blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using Worst Fit"

Enter Number of Blocks: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
22
2

Process No      Process Size   Block No
1               22             Not Allocated
2               2              1

Worst Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using Worst Fit
```

OUTPUT :



```
(kali@YUVARAJK)-[~]
$ nano wf.c
(kali@YUVARAJK)-[~]
$ gcc wf.c -o wf
(kali@YUVARAJK)-[~]
$ ./wf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
22
2

Process No      Process Size   Block No
1               22             Not Allocated
2               2              1

(kali@YUVARAJK)-[~]
$ nano wf.sh
(kali@YUVARAJK)-[~]
$ bash wf.sh
Worst Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using Worst Fit
(kali@YUVARAJK)-[~]
$
```